Algorithm Description Document

F-16 Missile Evasion: Defensive Break Turn Against AIM-9 Sidewinder Engagement

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1 Document Control

1.1 Revision History

Rev	Date	Description	Author
A	2026-03-17	Initial release — auto-generated from STELLA bundle `evasion.stella`	STELLA Agent

1.2 Applicable Documents

ID	Title	Version
AD-01	STELLA Framework Architecture Description	v0.2.5
AD-02	stellarust-gnc: Augmented Proportional Navigation Reference	v0.2.5
AD-03	stellarust-sim: Simulation Harness API	v0.2.5
AD-04	[MANUAL: F-16 flight dynamics data source]	
AD-05	[MANUAL: AIM-9 Sidewinder performance reference]	

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2 Introduction

2.1 Purpose

This document describes the guidance, navigation, and control (GNC) algorithms implemented in the STELLA simulation framework for the *F-16 defensive break turn missile evasion* scenario. It provides the mathematical basis, input/output interface definition, mode logic, and simulation-derived performance characterization for both the evading fighter aircraft (F-16 Fighting Falcon) and the pursuing missile (AIM-9 Sidewinder class).

The scenario evaluates whether a maximum-performance defensive break turn can defeat an infrared-guided short-range air-to-air missile launched from the rear hemisphere.

2.2 Scope

This ADD covers the following algorithm functions:

- **F-16 Flight Control:** Cruise, Defensive Break Turn (high-G maneuver), and Extend phases
- **F-16 Aerodynamic Model:** Lift/drag polar with structural G-limiting
- **Missile Guidance:** Augmented Proportional Navigation (APN) with gimbal-steered thrust
- **Missile Flight Phases:** Boost, Terminal, and Expended states
- **Engagement Assessment:** Range computation, closest point of approach (CPA), hit/miss determination

The following are explicitly *out of scope*: [MANUAL: Electronic countermeasures (ECM/ECCM), multi-aircraft tactics, weapon system integration beyond kinematic simulation]

2.3 Definitions and Acronyms

Term	Definition
ADD	Algorithm Description Document
APN	Augmented Proportional Navigation
CPA	Closest Point of Approach
FSW	Flight Software
GNC	Guidance, Navigation, and Control
GPS	Global Positioning System
IMU	Inertial Measurement Unit
IR	Infrared
LOS	Line of Sight
NED	North-East-Down coordinate frame
PD	Proportional-Derivative (controller)
PID	Proportional-Integral-Derivative (controller)
PN	Proportional Navigation
STELLA	Simulation Tool for Engagement-Level Lethality Analysis

3 System Overview

3.1 Engagement Configuration

This simulation models a tail-aspect infrared missile engagement against a maneuvering fighter aircraft in the NED (North-East-Down) reference frame. The F-16 detects the inbound missile via onboard radar and initiates a maximum-performance defensive break turn to defeat the missile's guidance solution.

3.1.1 F-16 Fighting Falcon

Parameter	Value
Vehicle ID	`f16`
Gross Mass	12,000 kg (combat weight)
Dry Mass	9,000 kg
Fuel Mass	3,000 kg
Engine Thrust	76,000 N (F110-GE-100, military power)
Wing Area	27.87 m^2
Moments of Inertia	$I_{xx} = 12,875$, $I_{yy} = 75,674$, $I_{zz} = 85,552$ kg $\cdot$ m^2
Initial Speed	300 m/s (Mach 0.88)
Initial Altitude	5,000 m MSL
Initial Heading	North (NED +X)
Structural G Limit	9.0 G
Sensors	IMU, GPS, Star Tracker, Radar (tracking missile)

3.1.2 AIM-9 Sidewinder Missile

Parameter	Value
Vehicle ID	`missile`
Dry Mass	55 kg
Fuel Mass	35 kg
Total Mass at Launch	90 kg
Motor Thrust	12,700 N
Specific Impulse	250 s
Estimated Burn Time	5.8 s
Estimated Delta-V	1,207 m/s
Max Lateral Accel	30 G (294.2 m/s^2)
Moments of Inertia	$I_{xx} = 0.5$, $I_{yy} = 15$, $I_{zz} = 15$ kg $\cdot$ m^2
Initial Speed	350 m/s

Launch Offset	3,000 m aft, 500 m east of F-16
Initial Range	3,041 m
Sensors	IMU, GPS, Star Tracker, Radar (tracking F-16)

3.2 GNC Architecture

Both vehicles employ independent flight software (FSW) modules operating at 100 Hz, with physics integrated at 1 kHz. Each FSW implements a phase-based finite state machine.

3.2.1 F-16 Phase State Machine

The F-16 FSW operates in three sequential phases:

1. **Cruise** (Phase 0): Level flight at 300 m/s, altitude hold via PD pitch control. Transitions to Break Turn upon radar detection of missile at range $< 3{,}500$ m or $t > 0.5$ s.
2. **Break Turn** (Phase 1): Maximum-performance right bank turn at 80° bank angle with aggressive angle-of-attack (13°). Duration: 12 seconds. Designed to force high crossing angle against the missile.
3. **Extend** (Phase 2): Roll wings level and accelerate away from the engagement area with 85% throttle.

3.2.2 Missile Phase State Machine

The missile FSW operates in three phases:

1. **Boost** (Phase 0): Full-thrust motor burn with gimbal-steered thrust vector for guidance corrections. Duration: 5.8 s.
2. **Terminal** (Phase 1): Motor burnout; guidance continues via residual thrust (if fuel remains) and aerodynamic steering torques. Transitions to Expended if closing speed drops below -100 m/s after $t > 10$ s.
3. **Expended** (Phase 2): Guidance ceases; missile is ballistic.

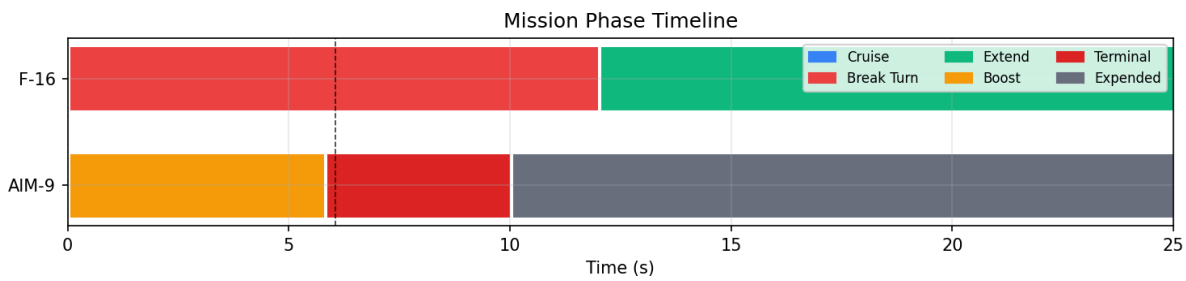

Figure 1: Mission Phase Timeline — F-16 and AIM-9 Sidewinder

4 F-16 Flight Control Algorithms

4.1 Aerodynamic Force Model

The F-16 aerodynamic model computes lift and drag forces in the body frame, then rotates them to the NED frame for integration.

4.1.1 Atmospheric Model

Density is computed using an exponential atmosphere:

$$\rho(h) = \rho_0 \exp\left(-\frac{h}{H}\right)$$

where $\rho_0 = 1.225$ kg/m³ (sea-level density) and $H = 8500$ m (scale height).

4.1.2 Lift and Drag

Dynamic pressure:

$$q = \frac{1}{2}\rho V^2$$

Lift coefficient (linear model, clamped at stall):

$$C_L = \text{clamp}\big(C_{L\alpha} \cdot \alpha, -C_{L,\max}, C_{L,\max}\big)$$

Drag coefficient (parabolic polar):

$$C_D = C_{D0} + \frac{C_L^2}{\pi e \ \text{AR}}$$

where:

Parameter	Value	Units
$C_{L\alpha}$	3.44	1/rad
C_{D0}	0.025	—
$C_{L,\max}$	1.2	—
Aspect Ratio (AR)	3.2	—
Oswald factor (e)	0.8	—
Wing area (S)	27.87	m²
Structural G limit	9.0	G

Lift force is clamped to enforce the structural G limit:

$$L_{\max} = n_{\text{limit}} \cdot m \cdot g_0 = 9.0 \times 12000 \times 9.807 = 1.059 \times 10^6 \text{ N}$$

4.1.3 Body-to-NED Rotation

Forces are computed in the body frame as $\boldsymbol{F}_{\text{body}} = [-D, 0, -L]^T$ and rotated to NED via the attitude quaternion:

$$\boldsymbol{F}_{\text{NED}} = \boldsymbol{q} \circ \boldsymbol{F}_{\text{body}} \circ \boldsymbol{q}^*$$

4.2 Attitude Control

4.2.1 Control Law

The F-16 employs PD attitude controllers for roll and pitch axes with yaw rate damping:

$$\tau_{\text{roll}} = K_{p,\text{roll}}(\varphi_{\text{cmd}} - \varphi) - K_{d,\text{roll}}\omega_x$$

$$\tau_{\text{pitch}} = K_{p,\text{pitch}}(\theta_{\text{cmd}} - \theta) - K_{d,\text{pitch}}\omega_y$$

$$\tau_{\text{yaw}} = -K_{d,\text{yaw}}\omega_z$$

4.2.2 Controller Gains

Parameter	Value	Units
$K_{p,\text{roll}}$	25,000	N · m/rad
$K_{d,\text{roll}}$	25,000	N · m · s/rad
$K_{p,\text{pitch}}$	150,000	N · m/rad
$K_{d,\text{pitch}}$	100,000	N · m · s/rad
$K_{d,\text{yaw}}$	30,000	N · m · s/rad
Max torque (all axes)	200,000	N · m

4.2.3 Phase-Specific Commands

Phase	Bank Cmd	Pitch Cmd	Thrust
Cruise	0°	2.9° + alt hold	40% (30,400 N)
Break Turn	80°	13.2° + alt correction	95% (72,200 N)
Extend	0°	3.4° + alt hold	85% (64,600 N)

4.2.4 Simulation Results — F-16 Flight Parameters

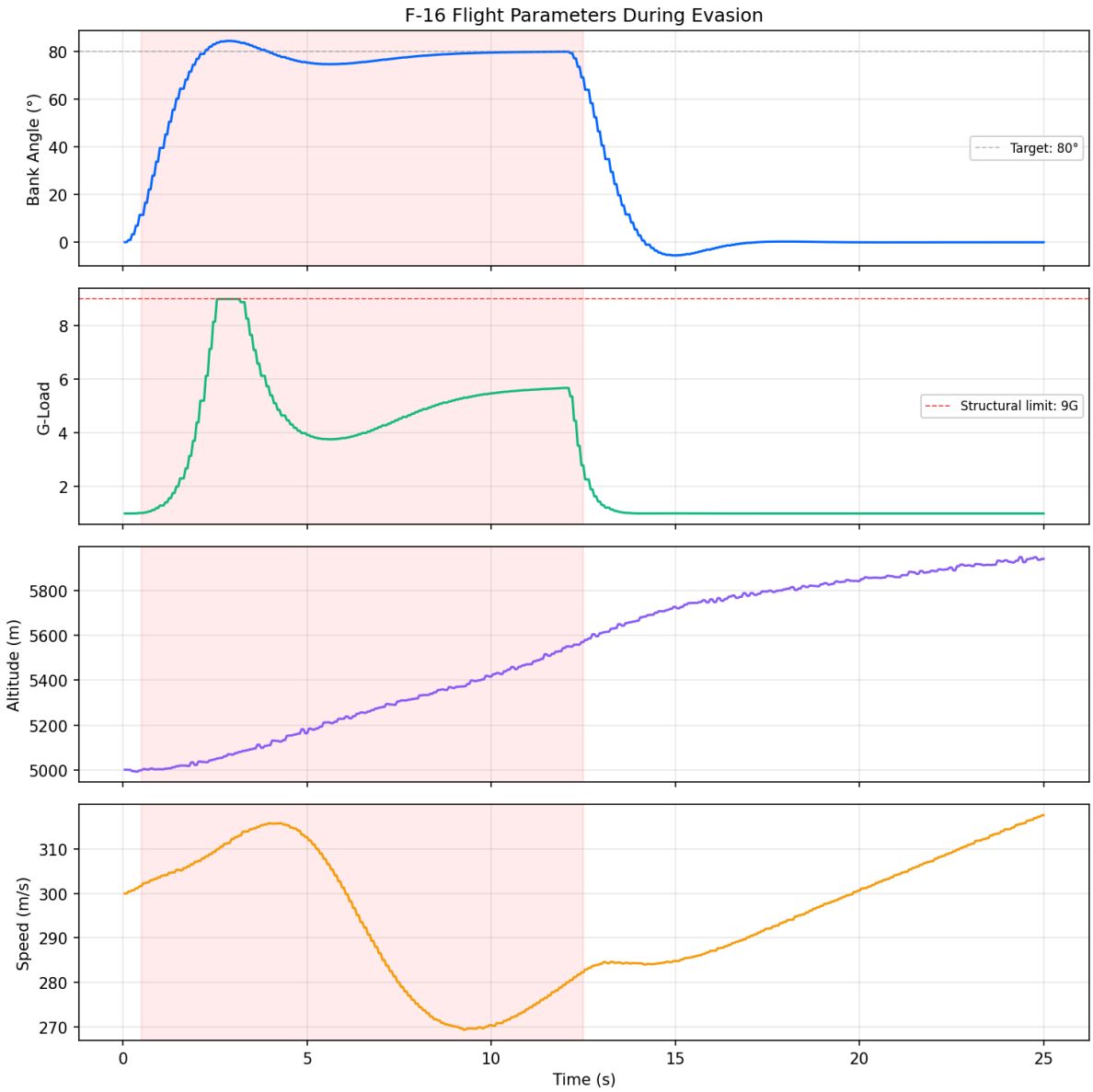

Figure 2: F-16 Flight Parameters During Evasion Maneuver

Metric	Value	Notes
Max bank angle	84.4°	Target: 80°; slight overshoot
Max G-load	9.0 G	At structural limit
Altitude range	4,993 – 5,948 m	Gained 950 m during break
Speed range	269 – 318 m/s	Bled 31 m/s in turn
Break turn duration	12.0 s	Commanded
Break turn start	$t = 0.05$ s	Immediate upon radar contact

5 Missile Guidance Algorithms

5.1 Augmented Proportional Navigation

5.1.1 Algorithm Description

The missile employs Augmented Proportional Navigation (APN), a standard homing guidance law that commands lateral acceleration proportional to the line-of-sight (LOS) rotation rate and closing velocity:

$$\boldsymbol{a}_{\text{cmd}} = N' \cdot V_c \cdot \boldsymbol{\Omega}_{\text{LOS}}$$

where N' is the effective navigation constant, V_c is the closing velocity (range rate), and $\boldsymbol{\Omega}_{\text{LOS}}$ is the angular rate of the LOS vector.

5.1.2 Guidance Parameters

Parameter	Value	Units
Navigation constant (N')	4.0	—
Max lateral acceleration	294.2 (30 G)	m/s^2
Max axial acceleration	50.0	m/s^2
Min active range	10.0	m
Adaptive mode	Off	—

5.2 Steering Implementation

5.2.1 Velocity Vector Tracking

The missile body tracks its velocity vector via PD torque control, preventing body wobble from rapid reorientation. The velocity direction in body frame is computed and errors around pitch (Y) and yaw (Z) axes are driven to zero:

$$\tau_{\text{pitch}} = k_p\left(-v_{z,\text{body}}\right) - k_d\omega_y$$

$$\tau_{\text{yaw}} = k_p\left(v_{y,\text{body}}\right) - k_d\omega_z$$

$$\tau_{\text{roll}} = -k_{d,\text{roll}}\omega_x$$

Parameter	Value	Units
k_p (steering)	100	N · m/rad
k_d (steering)	30	N · m · s/rad
$k_{d,\text{roll}}$	20	N · m · s/rad

5.2.2 Thrust Gimbal

During boost phase, guidance corrections are applied via thrust vector control (gimbal pitch and yaw), with gimbal limits of ±0.3 rad (17.2°) during boost and ±0.5 rad (28.6°) during terminal phase. The gimbal angles are computed by projecting the desired acceleration vector into body frame:

$$\delta_{\text{pitch}} = \text{clamp}\big(\text{atan2}\big(-a_{z,\text{body}}, a_{x,\text{body}}\big), -\delta_{\max}, \delta_{\max}\big)$$

$$\delta_{\text{yaw}} = \text{clamp}\big(\text{atan2}\big(a_{y,\text{body}}, a_{x,\text{body}}\big), -\delta_{\max}, \delta_{\max}\big)$$

5.3 Target Tracking

The missile reconstructs target position from radar measurements (range, azimuth, elevation) and estimates target velocity from successive position updates:

$$\boldsymbol{r}_{\text{target}} = \boldsymbol{r}_{\text{own}} + \hat{\boldsymbol{u}}_{\text{LOS}} \cdot R$$

$$\boldsymbol{v}_{\text{target}} \approx \frac{\boldsymbol{r}_{\text{target}(t)} - \boldsymbol{r}_{\text{target}(t-\Delta t)}}{\Delta} t$$

5.4 Flight Phase Performance

5.4.1 Phase Timing

Phase	Start	End	Duration
Boost	$t = 0.05$ s	$t = 5.85$ s	5.80 s
Terminal	$t = 5.85$ s	$t = 10.05$ s	4.20 s
Expended	$t = 10.05$ s	$t = 25.0$ s	14.95 s

5.4.2 Missile Guidance Telemetry

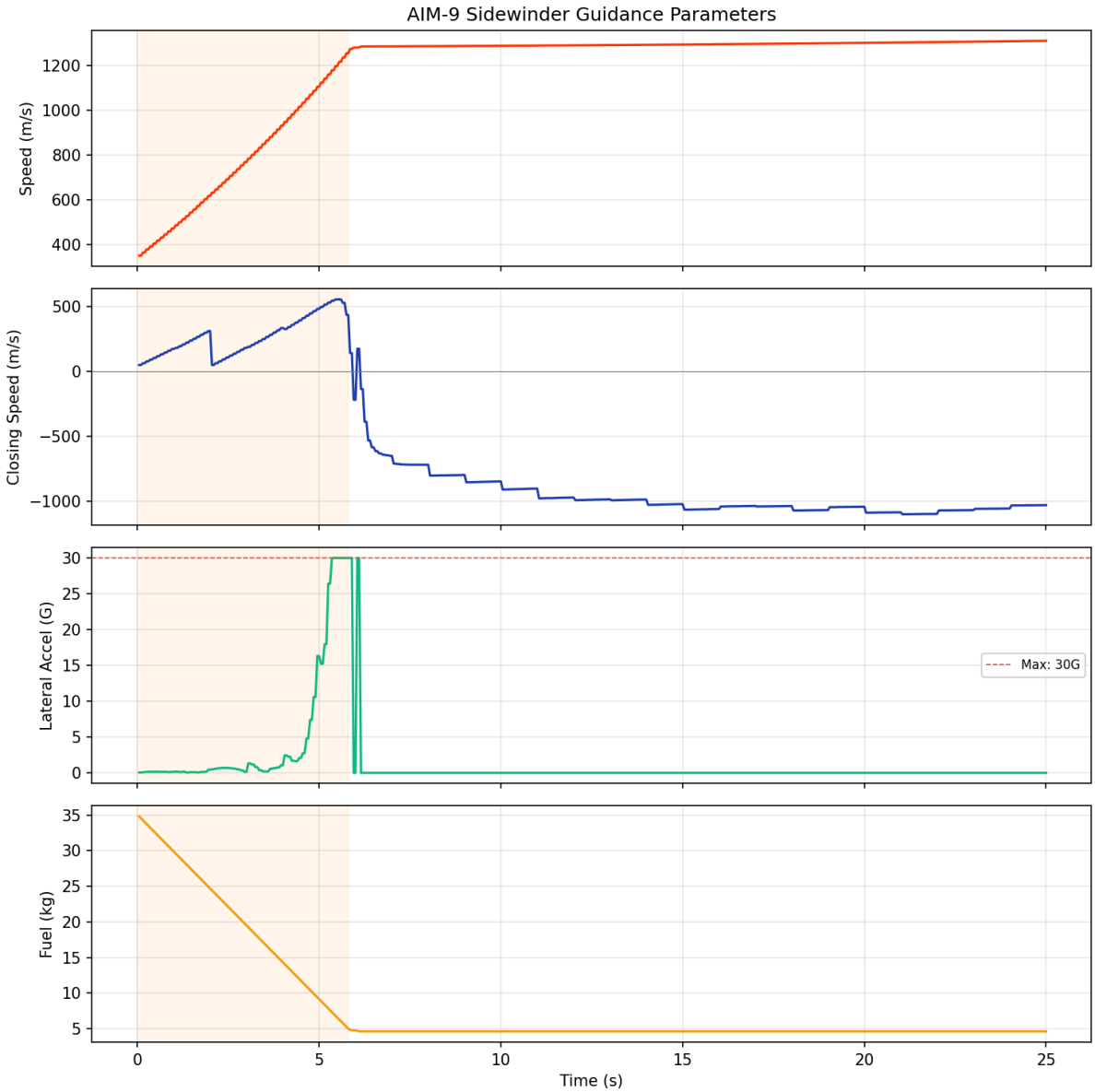

Figure 3: AIM-9 Sidewinder Guidance Parameters During Engagement

Metric	Value	Notes

Peak speed	1,310 m/s	Mach 3.9 at end of boost
Max closing speed	556 m/s	During initial approach
Max lateral accel	30.0 G	At guidance limit
Final fuel remaining	4.6 kg	Terminal phase used residual thrust
Phase at simulation end	Expended	Diverging from target

6 Engagement Analysis

6.1 Engagement Geometry

The missile launches from the rear quarter of the F-16 (3,000 m aft, 500 m east) at $t = 0$. The F-16 immediately initiates a hard right break turn at 80° bank, pulling 9 G. This forces a rapidly increasing crossing angle, driving up the LOS rotation rate faster than the missile's lateral acceleration can track.

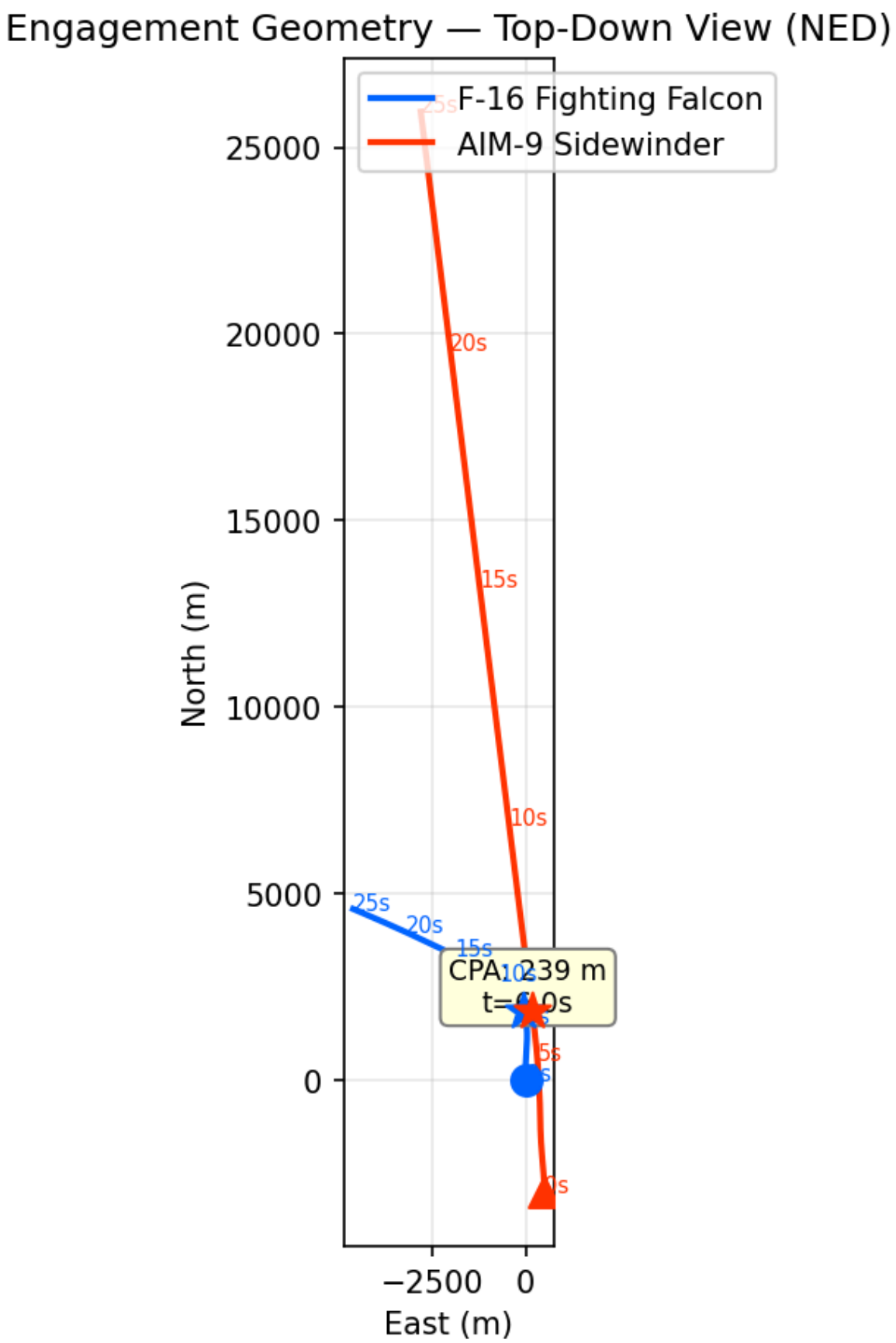

Figure 4: Engagement Geometry — Top-Down View (NED Frame)

6.2 Range Profile

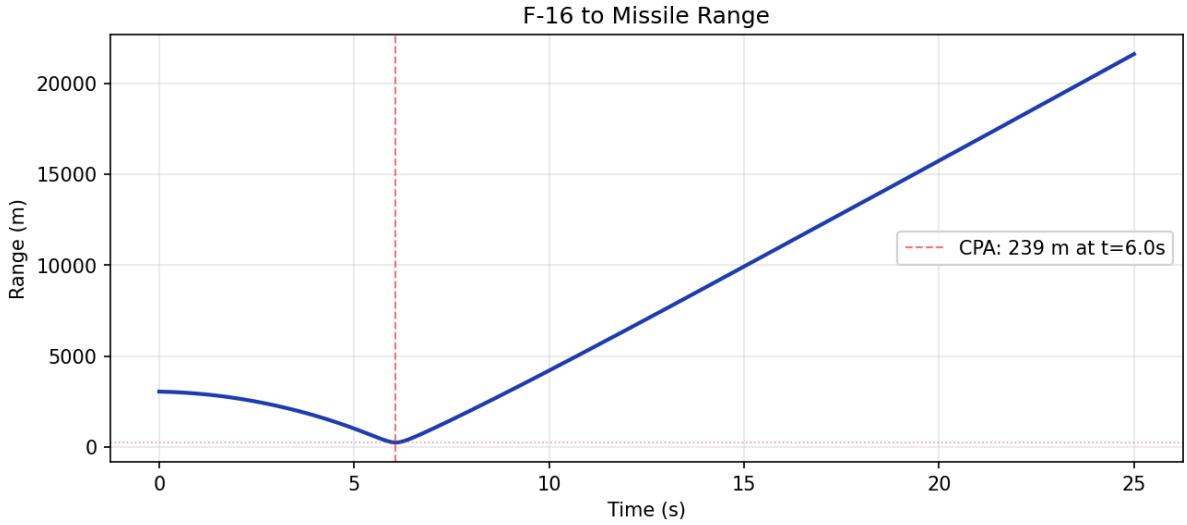

Figure 5: F-16 to Missile Range vs. Time

The range decreases from 3,041 m at $t = 0$ to a minimum of **239.1 m** at $t = 6.05$ s (closest point of approach). After CPA, the range monotonically increases as the missile's energy is insufficient to re-engage.

6.3 Closing Geometry

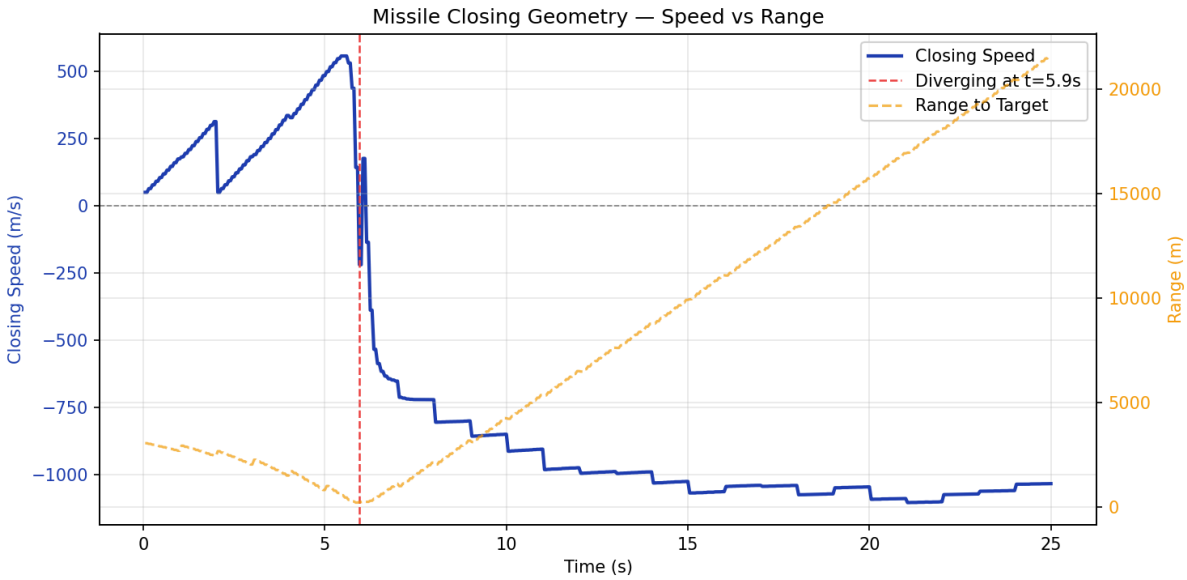

Figure 6: Missile Closing Speed and Range to Target

6.4 3D Trajectory

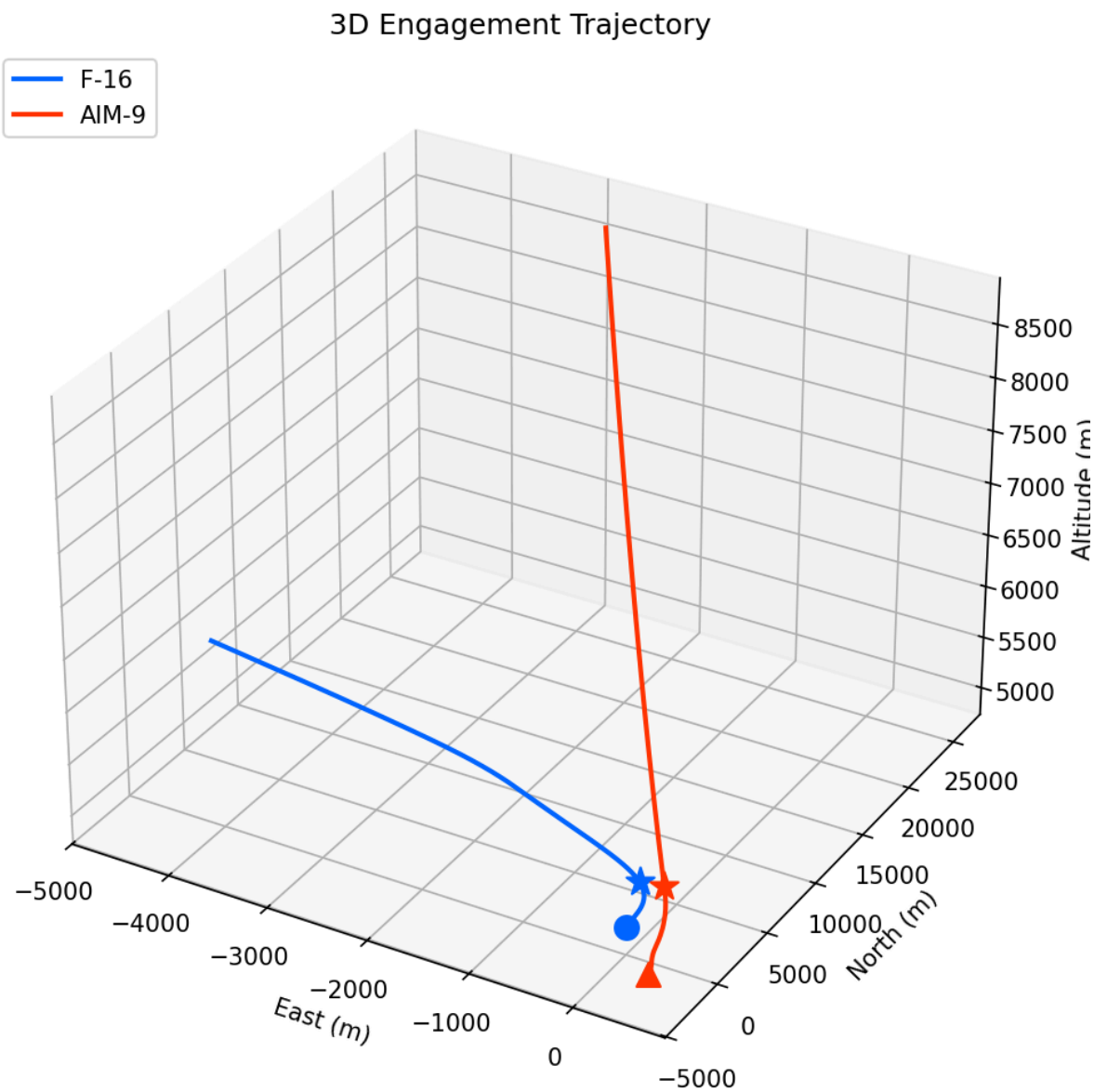

Figure 7: 3D Engagement Trajectory — F-16 and AIM-9 Sidewinder

6.5 Altitude Profiles

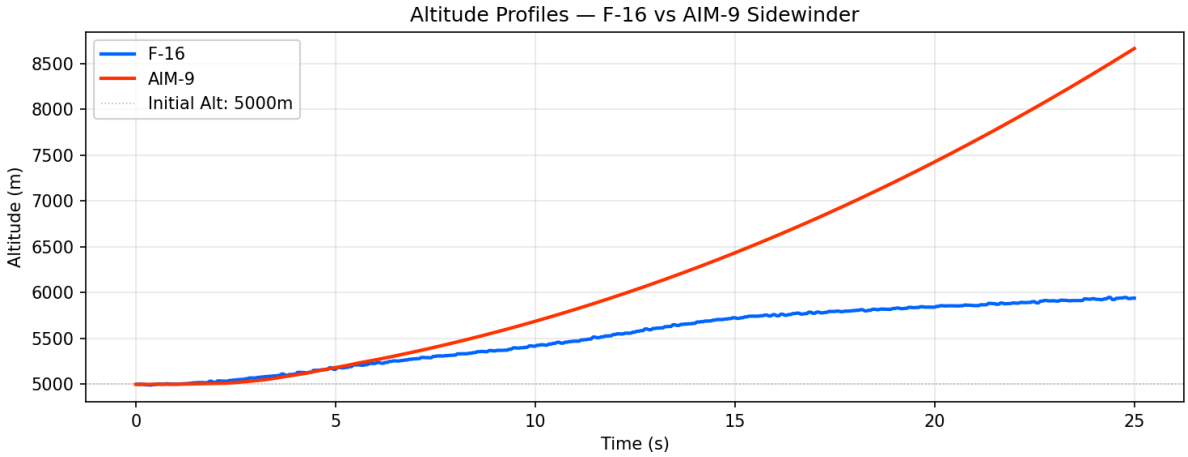

Figure 8: Altitude Profiles — F-16 vs AIM-9 Sidewinder

The F-16′s break turn causes a climb from 5,000 m to approximately 5,950 m as the high bank angle and aggressive pitch command produce a vertical lift component. The missile initially tracks at the same altitude but diverges as it expends energy attempting to match the F-16′s lateral maneuver.

6.6 Engagement Summary

Metric	Value
Outcome	**MISSILE MISS — F-16 survived**
Miss distance (CPA)	239.1 m
Time of CPA	$t = 6.05$ s
Kill radius	5.0 m
Miss margin	234.1 m (47× kill radius)
Initial range	3,041 m
Simulation duration	25.0 s
F-16 final state	Wings level, extending at 285 m/s
Missile final state	Expended, diverging

7 Mission Timeline and Mode Transitions

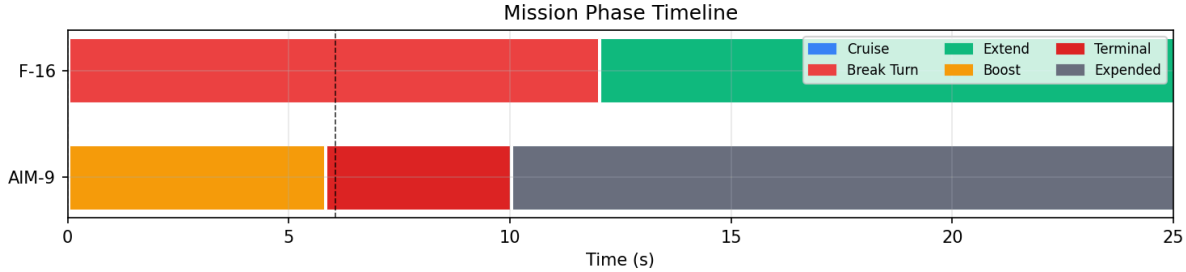

Figure 9: GNC Mode Transition Timeline — All Vehicles

T (s)	Vehicle	Event / Transition	Notes
0.00	Both	Simulation Start	Engagement initiated
0.05	F-16	Cruise → Break Turn	Radar detects missile
0.05	Missile	Boost phase begins	Motor ignition
5.85	Missile	Boost → Terminal	Motor burnout; 4.6 kg fuel remaining
6.05	F-16	Closest Approach: 239 m	CPA — missile misses
6.05	Missile	MISSILE MISS	Beyond kill radius (5 m)
10.05	Missile	Terminal → Expended	Closing speed < -100 m/s; guidance ceases
12.05	F-16	Break Turn → Extend	12 s turn complete; wings level
25.00	Both	Simulation End	

8 Error Budget and Sensitivity Analysis

8.1 Sensor Configuration

Both vehicles use default STELLA sensor configurations:

Sensor	Configuration	Notes
IMU	Default ImuConfig	Gyro + accelerometer noise
GPS	Default GpsConfig	Position + velocity measurements
Star Tracker	Default StarTrackerConfig	Attitude quaternion measurement
Radar	Default RadarConfig	Range, range-rate, azimuth, elevation

8.2 Key Sensitivities

The engagement outcome (miss distance) is sensitive to the following parameters. A full Monte Carlo sensitivity analysis is recommended (see Section 9):

Parameter	Expected Impact	Priority
F-16 break turn timing	Earlier break increases miss; delayed break decreases miss	High
F-16 bank angle command	Higher bank = higher crossing rate = larger miss	High
Missile navigation constant (N')	Higher N' improves tracking of maneuvering targets	High
Missile max lateral G	Higher G limit allows tighter turns	High
Initial range	Shorter range gives missile less time to lose energy	Medium
F-16 speed at break initiation	Higher speed = tighter turn at same G	Medium
Radar noise / latency	Degrades missile target estimation	Low

8.3 Gravity Model

Both vehicles use a flat-gravity model ($g = 9.80665$ m/s^2, downward) appropriate for the engagement altitude range (5,000–6,000 m) and short duration (25 s). No J2, EGM2008, or centrifugal corrections are applied.

9 Verification and Validation

9.1 Simulation Configuration

Parameter	Value
STELLA version	stellarust v0.2.5
Physics rate	1,000 Hz (dt = 0.001 s)
FSW rate	100 Hz (dt = 0.01 s)
Telemetry record rate	20 Hz (interval = 0.05 s)
Simulation duration	25.0 s
Number of vehicles	2 (F-16, AIM-9)
Coordinate frame	NED (North-East-Down)
Bundle format	v2 (.stella directory with parquet)
Bundle ID	`evasion.stella`

9.2 Test Cases

ID	Description	Expected	Result
TC-01	Missile misses F-16 (CPA > kill radius)	CPA > 5 m	**PASS** (239.1 m)
TC-02	F-16 achieves target bank angle	Bank ≥ 75°	**PASS** (84.4°)
TC-03	F-16 G-load within structural limit	G ≤ 9.0	**PASS** (9.0 G)
TC-04	F-16 maintains positive altitude	Alt > 0 m	**PASS** (min 4,993 m)
TC-05	Missile achieves boost phase burnout	Burn ≈ 5.8 s	**PASS** (5.80 s)
TC-06	Missile guidance saturates (max lateral G)	Lat G = 30	**PASS** (30.0 G observed)
TC-07	Missile transitions to Expended on divergence	Phase = 2 after diverge	**PASS** ($t = 10.05$ s)
TC-08	Simulation completes without numerical errors	No NaN/Inf	**PASS**

10 Known Limitations and Open Items

ID	Description	Status
LIM-01	Single-run deterministic simulation — Monte Carlo analysis pending for statistical confidence on miss distance	Open
LIM-02	Flat gravity model (g = const) — acceptable for short-duration, low-altitude engagement	Accepted
LIM-03	Exponential atmosphere model — NRLMSISE-00 or US Standard Atmosphere not implemented	Open
LIM-04	No aerodynamic drag model on missile — drag forces not modeled, only thrust and gravity	Open
LIM-05	Simplified F-16 aero: no sideslip, no compressibility effects, no control surface deflection limits	Open
LIM-06	Star tracker sensor unrealistic for air-to-air scenario — used as generic attitude sensor	Accepted
LIM-07	Radar model provides perfect azimuth/elevation — no angle noise or track jitter	Open
LIM-08	No missile seeker model (IR/RF) — radar-based tracking instead of realistic IR seeker cone	Open
LIM-09	No pilot model — F-16 FSW executes pre-programmed maneuver without decision logic	Accepted
LIM-10	First telemetry sample contains NaN for some channels (initialization artifact)	Accepted

11 References

1. Zarchan, P., *Tactical and Strategic Missile Guidance*, 7th ed., AIAA, 2019.
2. Stevens, B.L. and Lewis, F.L., *Aircraft Control and Simulation*, 3rd ed., Wiley, 2015.
3. Siouris, G.M., *Missile Guidance and Control Systems*, Springer, 2004.
4. Yanushevsky, R., *Modern Missile Guidance*, CRC Press, 2008.
5. [MANUAL: F-16 flight performance data source / NATOPS equivalent]
6. [MANUAL: AIM-9 Sidewinder unclassified performance reference]
7. [MANUAL: Program-specific requirements documents]

12 Appendix A: Telemetry Channel Definitions

12.1 F-16 Channels

Channel	Units	Description
`f16_phase`	—	FSW phase: 0=Cruise, 1=BreakTurn, 2=Extend
`f16_g_load`	G	Normal load factor from bank angle: $\frac{1}{\cos(\varphi)}$
`f16_altitude_m`	m	Altitude above sea level ($-z$ in NED)
`f16_speed_mps`	m/s	Total airspeed (velocity magnitude)
`f16_bank_angle_deg`	deg	Roll angle extracted from attitude quaternion
`range_to_missile`	m	Radar-measured range to missile entity

12.2 Missile Channels

Channel	Units	Description
`missile_phase`	—	FSW phase: 0=Boost, 1=Terminal, 2=Expended
`missile_range_to_target`	m	Computed range to F-16
`missile_closing_speed`	m/s	Range rate (positive = closing)
`missile_time_to_go`	s	Estimated time to intercept from guidance
`missile_lateral_accel_g`	G	Commanded lateral acceleration magnitude
`missile_speed_mps`	m/s	Total missile speed
`missile_fuel_kg`	kg	Remaining propellant mass

13 Appendix B: Engagement Geometry Detail

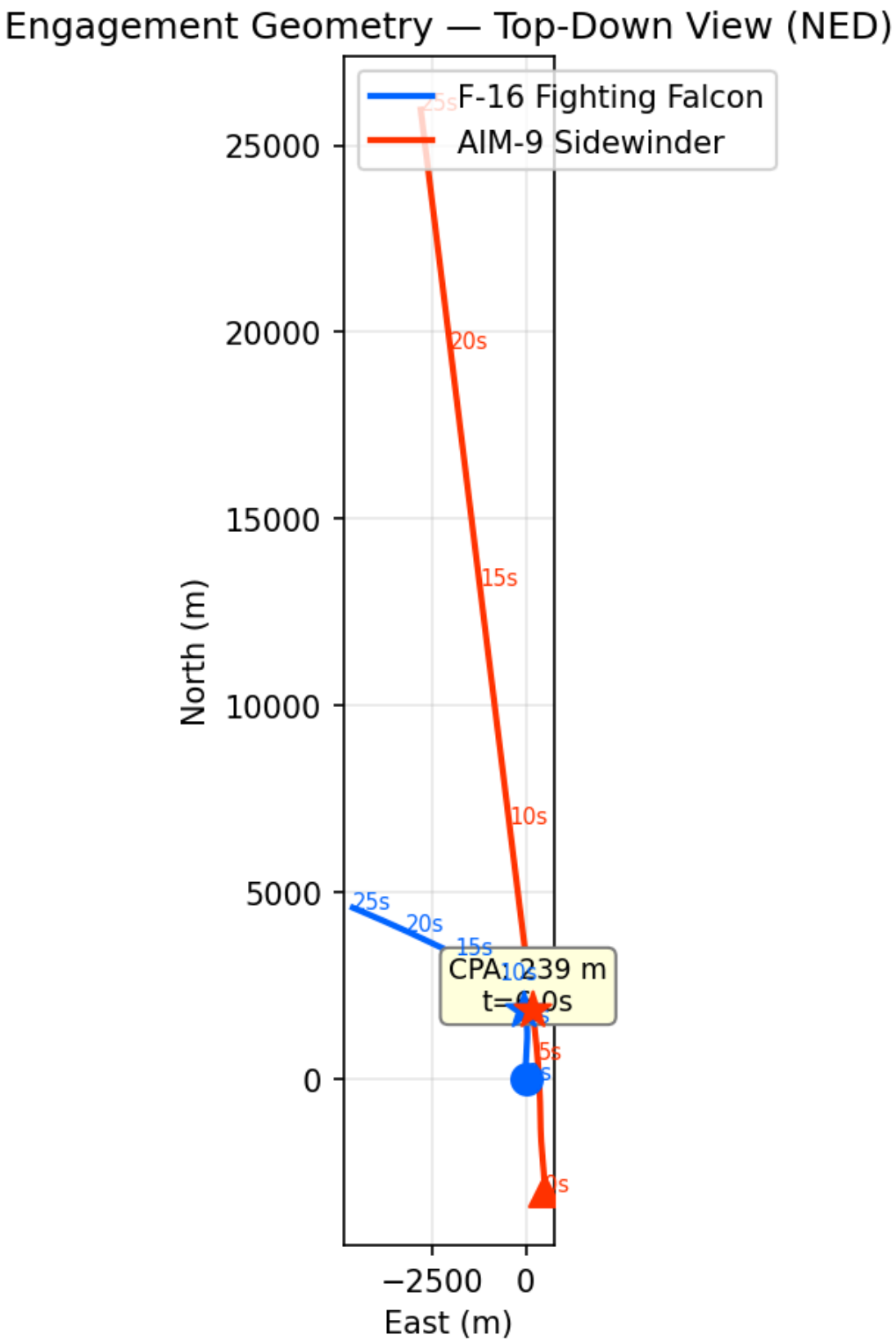

Figure 10: Top-down engagement geometry with time annotations

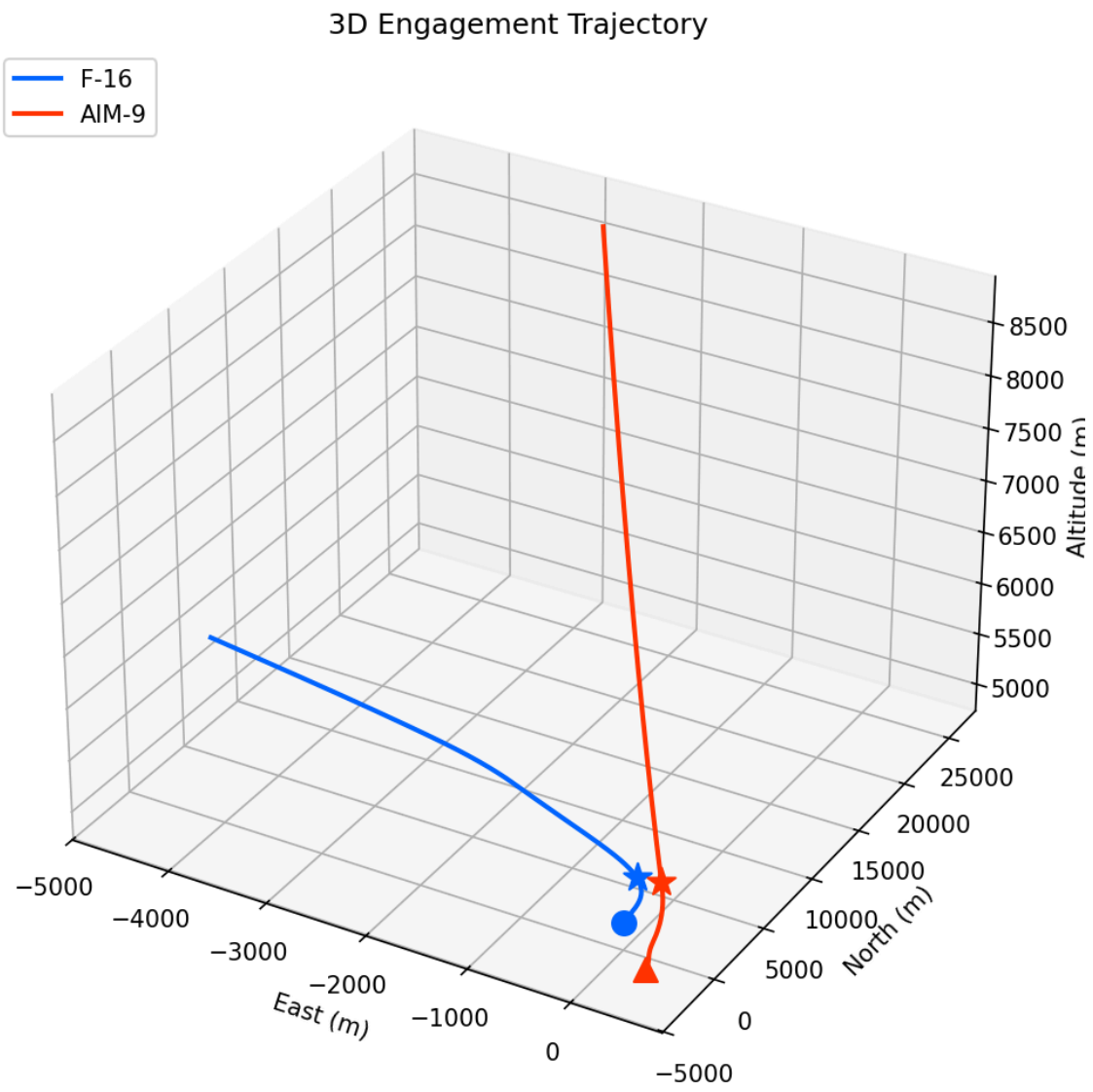

Figure 11: 3D trajectory showing altitude separation